

EXPERIMENT NO. 10

Prepare RMMM Plan for the Project

Project: Hospital Management System (HMS)

1. Aim

To identify risks and prepare the Risk Mitigation, Monitoring, and Management (RMMM) Plan for the Hospital Management System (HMS) project.

2. Objectives

- To identify potential technical, project, and business risks for the HMS project.
- To assess the probability and impact of each risk.
- To define mitigation, monitoring, and management strategies for the most significant risks.

3. Software Required

Any text editor / open source tool.

4. Risk Table

Risk	Category	Probability	Impact	Mitigation Strategy
Loss of patient data due to server/database failure	Technical	Low	Catastrophic	Daily automated backups; RAID storage; cloud-based redundant DB replication.
Unauthorized access to sensitive patient records	Technical	Medium	Catastrophic	Role-based access control, encryption, audit logs, regular security testing.
Payment gateway integration failure/downtime	Technical	Medium	Critical	Use a well-tested third-party gateway; implement retry & fallback (manual/cash) payment mode.
Requirement changes from hospital administration mid-project	Project	Medium	Marginal	Freeze core requirements after sign-off; handle later changes via a formal change-request process.
Delay in module integration (Patient, Billing, Pharmacy, Lab)	Project	Medium	Critical	Follow the V-Model schedule strictly; conduct integration testing incrementally, not only at the end.
Team member unavailability (illness, exams, etc.)	Project	Medium	Marginal	Cross-train team members; maintain shared documentation so any member can continue a module.
Low user adoption by hospital staff (resistance to new system)	Business	Low	Critical	Conduct training sessions; design a simple, intuitive UI; phased rollout with pilot department.
Underestimated effort for UML design and testing phases	Project	Medium	Marginal	Add buffer time in the Gantt chart; prioritize critical-path activities identified via CPM.

5. RMMM Plan (Selected Key Risks)

For the risks with the highest probability x impact, a specific monitoring approach and a management/contingency trigger are defined below.

ID	Risk	Monitoring Approach	Management / Trigger
RE-1	Payment gateway integration failure/downtime	Monitor gateway uptime/response time during each testing cycle; log failed transaction rate.	If failure rate > 5% in a testing cycle, trigger fallback to manual/offline payment recording.
RE-2	Delay in module integration	Track actual vs planned progress weekly against the Gantt chart / critical path (CPM).	If a critical-path activity slips by more than 2 days, reallocate resources from non-critical (slack) activities.
RE-3	Unauthorized access to patient records	Review access logs weekly; run periodic vulnerability scans on the login/authentication module.	If a breach or vulnerability is found, patch immediately, force password resets, and notify hospital admin.
RE-4	Requirement changes from hospital administration	Log every new change request against the frozen SRS baseline.	If a change significantly affects scope, escalate through the formal change-request process and re-baseline the schedule.

6. Conclusion

The Risk Table identifies eight technical, project, and business risks relevant to the HMS project, ranked by probability and impact. The RMMM Plan defines concrete mitigation steps, monitoring approaches, and contingency triggers for the four most significant risks - payment gateway failure, integration delay, unauthorized data access, and scope changes - helping the team respond proactively rather than reactively as the project progresses through the V-Model.